

Abhinav Raj

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EDUCATION

University of Illinois Urbana-Champaign

B.S. in Computer Science and Statistics; Minor in Mathematics

Champaign, IL

Expected May 2029

- GPA: 4.0/4.0; Dean's List.
- Coursework: Data Structures and Algorithms (in progress), Computer Architecture (in progress), Discrete Structures, Calculus III, Linear Algebra, Probability and Statistics.

TECHNICAL SKILLS

- **Languages/Tools:** Rust, C++, Java, Python (NumPy, Pandas, Matplotlib), SQL, Git.

EXPERIENCE

Financial Engineering Club

Member

Champaign, IL

Feb 2026 – Present

- Built a Python research pipeline comparing Buy and Hold, 252-day Time-Series Momentum, Directional Change, and Multi-Threshold Directional Change strategies on SPY, QQQ, TLT, and GLD daily ETF data from 2010 to 2026.
- Implemented one-day-lagged backtests with normalized exposures, turnover-based transaction costs, and 0, 5, and 10 bps robustness checks to evaluate sensitivity to trading costs.
- Tracked return, Sharpe ratio, volatility, max drawdown, turnover, and trade counts across strategies to compare performance and risk-adjusted returns.
- Found Time-Series Momentum was the strongest active strategy, earning 76.62% total return with 0.41 Sharpe and 3.73% turnover, while higher-turnover directional-change variants underperformed after costs.

SIGNLL Research Group

Research Member

Champaign, IL

Aug 2025 – Present

- Benchmarked locally hosted 3B small language models, including Llama 3.2, Qwen 2.5, and Phi 3.5, against Claude Haiku 4.5 and Sonnet 4.6 on FinQA financial numerical reasoning.
- Measured quality, latency, and cost using multi-tolerance accuracy, bootstrap confidence intervals, and paired-bootstrap significance tests for reproducible model comparison.
- Identified a statistically robust 22.4-point accuracy gap between Haiku at 80.0% and the best small language model at 57.6%, while local models ran at zero marginal API cost.
- Built a unit-aware answer-extraction and dual-gold scoring layer with 99.6%+ parse coverage; validated that small language model errors were reasoning failures rather than parsing artifacts.

Solar Car Team Illinois

Strategy Team Member

Champaign, IL

Aug 2025 – Dec 2025

- Identified anomalous battery discharge patterns across 240K+ telemetry records using Python and SQL, informing race strategy adjustments and data-quality pipeline improvements.
- Parsed JSON logs and visualized trends with Pandas, Polars, NumPy, and Matplotlib to support telemetry analysis and data-quality improvements.

PROJECTS

NYC Traffic Fatality Prediction

June 2024

- Built an end-to-end Python classification workflow for preprocessing, feature encoding, resampling, model evaluation, and error analysis on NYC traffic accident data.
- Achieved a 0.87 F1-score on the minority fatality class across 74,881 traffic accident records using SMOTE oversampling and one-hot encoding in a highly imbalanced classification setting.
- Evaluated precision, recall, F1-score, and confusion matrix results; reviewed error slices by borough, time of day, and weather conditions.

SilentSync

Founder

2023 – 2025

- Launched an app to support deaf employees and improve workplace communication; led product strategy, user research, and product iteration.
- Conducted 100+ user interviews and incorporated feedback from deaf users and retail partners into product improvements.
- Secured local business partnerships, increased usage by 40%, and reached 150+ downloads in the first month.
- Published peer-reviewed research on deaf accessibility in IJHSS and presented at the Innosphere Research Conference.